GROUP (7) CHM PROJECT ASSIGNMENT ON EARTHQUAKE

EXTRACTION OF USEFUL INFORMATION FROM THE EARTHQUAKE TABLE

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-- 1. Get all earthquake records with selected fields

SELECT earthquake_id, occurred_on, latitude, longitude, depth, magnitude, calculation_method, network_id, place, cause

FROM earthquake_tb

LIMIT 16;

earthquake_id	occurred_on	latitude	longitude	depth	magnitude	calculation_method	network_id	place	cause
1	1969-01-01 09:07:00	51.096	-179.392	45	5.6	mw	iscgem812771	Andreanof Islands, Aleutian Islands, Alaska	earthquake
2	1969-01-02 17:50:00	-56.096	-27.842	80.1	6	mw	iscgemsup812819	South Sandwich Islands region	earthquake
3	1969-01-03 03:16:00	37.140	57.899	10	5.5	mw	iscgem812826	Turkmenistan-Iran border region	earthquake
4	1969-01-03 13:28:00	51.132	-179.306	15	5.9	mw	iscgem812841	Andreanof Islands, Aleutian Islands, Alaska	earthquake
5	1969-01-04 22:36:00	-6.850	129.821	105	5.8	mw	iscgem812879	Banda Sea	earthquake
6	1969-01-05 13:26:00	-7.970	158.888	60	7	mw	iscgem812898	Solomon Islands	earthquake
7	1969-01-06 15:39:00	-10.407	164.479	10	6.9	mw	iscgem812954	Santa Cruz Islands region	earthquake
8	1969-01-08 21:55:00	11.823	143.419	45	5.6	mw	iscgem813024	south of the Mariana Islands	earthquake
9	1969-01-10 03:20:00	28.920	130.667	25	5.9	mw	iscgem813057	Ryukyu Islands, Japan	earthquake
10	1969-01-13 08:55:00	-7.967	158.999	50	5.8	mw	iscgem813149	Solomon Islands	earthquake
11	1969-01-14 23:12:00	36.043	29.221	25	6.4	mw	iscgem813187	western Turkey	earthquake
12	1969-01-18 03:02:00	-56.756	-26.865	130	6.4	mw	iscgemsup813263	South Sandwich Islands region	earthquake
13	1969-01-19 07:02:00	44.985	143.238	244.7	7	mw	iscgem813290	Hokkaido, Japan region	earthquake
14	1969-01-19 18:50:00	-14.849	167.345	112.9	7.1	mw	iscgem813304	Vanuatu	earthquake
15	1969-01-20 12:24:00	-10.441	164.67	10	6.2	mw	iscgem813332	Santa Cruz Islands region	earthquake
16	1969-01-20 14:20:00	54.899	166.054	25	5.9	mw	iscgem813336	Komandorskiye Ostrova, Russia region	earthquake

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-- The top 5 most recent earthquakes

SELECT earthquake_id, latitude, longitude, depth, magnitude, place, cause, occurred_on

FROM earthquake_tb

ORDER BY occurred_on DESC

LIMIT 5;

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: | Fetch rows:

earthquake_id	latitude	longitude	depth	magnitude	place	cause	occurred_on
23119	54.427	-161.5131	31	6	114km SE of Cold Bay, Alaska	earthquake	2018-12-31 02:35:00
23118	-2.677	102.349	166	5.7	89km NW of Lubuklinggau, Indonesia	earthquake	2018-12-30 08:39:00
23117	5.898	126.9209	60.21	7	96km ESE of Pondaguitan, Philippines	earthquake	2018-12-29 03:39:00
23116	-1.452	134.085	41	5.7	64km S of Manokwari, Indonesia	earthquake	2018-12-28 03:03:00
23115	-17.279	-174.0153	120	5.7	149km S of Hihifo, Tonga	earthquake	2018-12-26 14:11:00
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

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-- Find the earthquake with highest magnitude by country

SELECT place AS country, MAX(magnitude) AS highest_magnitude

FROM earthquake_tb

GROUP BY place

ORDER BY highest_magnitude DESC

LIMIT 1;

country	highest_magnitude
near the east coast of Honshu, Japan	9.1

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-- Find the top 5 regions with the most earthquakes

SELECT place, COUNT(*) AS Earthquake_Count

FROM earthquake_tb

GROUP BY place

ORDER BY Earthquake_Count DESC

LIMIT 5;

place	Earthquake_Count
Vanuatu	699
Tonga	632
Fiji region	632
Kuril Islands	584
New Britain region, Papua New Guinea	470

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-- Get earthquakes in a specific place (Tonga)

SELECT latitude, longitude, occurred_on, place, magnitude, depth

FROM earthquake_tb

WHERE place = 'Tonga';

latitude	longitude	occurred_on	place	magnitude	depth
-15.426	-174.909	1969-01-31 14:59:00	Tonga	5.6	260
-16.695	-174.666	1969-05-01 19:05:00	Tonga	6.2	203.4
-15.333	-173.165	1969-08-26 20:28:00	Tonga	5.6	35
-16.167	-173.704	1969-10-26 06:38:00	Tonga	5.8	115
-20.277	-173.861	1970-04-02 11:11:00	Tonga	5.9	15
-20.666	-173.859	1970-04-03 06:52:00	Tonga	6	25
-18.966	-173.545	1970-04-11 06:21:00	Tonga	5.8	25

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--- Get earthquakes with magnitude between 4.0 and 5.5

SELECT latitude, longitude, occurred_on, place, magnitude, depth

FROM earthquake_tb

WHERE magnitude BETWEEN 4.0 AND 5.5;

latitude	longitude	occurred_on	place	magnitude	depth
37.140	57.899	1969-01-03 03:16:00	Turkmenistan-Iran border region	5.5	10
76.481	131.227	1969-04-07 20:26:00	Laptev Sea	5.5	15
53.179	-160.38	1969-05-23 13:04:00	1969-04-07 20:26:00	5.5	15
-5.894	146.745	1969-06-24 03:29:00	eastern New Guinea region, Papua New Guinea	5.5	100
51.102	-179.142	1969-09-12 15:00:00	Andreanof Islands, Aleutian Islands, Alaska	5.5	35
51.827	175.366	1969-09-15 14:45:00	Rat Islands, Aleutian Islands, Alaska	5.5	33.5
45.532	151.738	1969-09-15 18:47:00	Kuril Islands	5.5	38.8

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— Get Earthquakes with lowest magnitude. Limit to 3

SELECT place AS country, MIN(magnitude) AS lowest_magnitude

FROM earthquake_tb

GROUP BY place

ORDER BY lowest_magnitude ASC

LIMIT 3;

country	lowest_magnitude
Andreanof Islands, Aleutian Islands, Alaska	5.5
South Sandwich Islands region	5.5
Turkmenistan-Iran border region	5.5

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-- Get earthquakes ordered by magnitude ascending. Limit to 1.

SELECT latitude, longitude, occurred_on, place, magnitude

FROM earthquake_tb

ORDER BY magnitude ASC

LIMIT 1;

latitude	longitude	occurred_on	place	magnitude
37.140	57.899	1969-01-03 03:16:00	Turkmenistan-Iran border region	5.5

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-- Get earthquakes ordered by depth ascending

SELECT latitude, longitude, occurred_on, place, cause, depth

FROM earthquake_tb

ORDER BY depth ASC;

latitude	longitude	occurred_on	place	cause	depth
34.131	-116.408	1992-06-28 12:00:00	3km NE of Yucca Valley, California	earthquake	-1.1
34.200	-116.437	1992-06-28 11:57:00	Landers, California Earthquake	earthquake	-0.097
37.621	-118.3425	1986-07-21 22:07:00	Central California	earthquake	-0.076
-21.812	-138.959	1987-11-05 17:29:00	Tuamotu Archipelago, French Polynesia region	nuclear explosion	0
49.917	78.945	1988-04-03 01:33:00	east Tuamotu Archipelago, French Polynesia region	explosion	0
-21.822	-139.029	1987-10-23 16:49:00	Tuamotu Archipelago, French Polynesia region	nuclear explosion	0
-22.252	-138.734	1989-06-10 17:29:00	Tuamotu Archipelago, French Polynesia region	nuclear explosion	0

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-- Get average magnitude of earthquakes per cause

SELECT cause, AVG(magnitude) AS avg_magnitude

FROM earthquake_tb

GROUP BY cause;

cause	avg_magnitude
earthquake	5.874561938802424
nuclear explosion	5.857803468208096
explosion	5.85